A Project Report

on

AI-DRIVEN EARLY SCREENING OF OBSTRUCTIVE SLEEP APNEA VIA SINGLE-LEAD ECG SIGNALS

Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY

Submitted by

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Major Project Phase - I

DEPARTMENT OF INFORMATION TECHNOLOGY

VNR Vignana Jyothi Institute of Engineering & Technology

(Autonomous Institute, Accredited by NAAC with 'A++' grade and NBA)

Bachupally, Nizampet (S.O.) Hyderabad- 500 090

November 2025

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Under the esteemed guidance of

PROJECT GUIDE

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Department of Information Technology

CERTIFICATE

This is to certify that the project work entitled "**AI-DRIVEN EARLY SCREENING OF OBSTRUCTIVE SLEEP APNEA VIA SINGLE-LEAD ECG SIGNALS**", being submitted by **A.Vasishta Varma (22071A1201), Chedapangu Anand (22071A1217), K. Harsha Vardhan (22071A1240), Madikuntawar Srikar (22071A1241)** in partial fulfillment for the award of Degree of **BACHELOR OF TECHNOLOGY** in **INFORMATION TECHNOLOGY** during the academic year **2025-26** is a record of bona-fide work carried out by them under our guidance and supervision.

The results embodied in this report have not been submitted by the students to any other University or Institution for the award of any degree or diploma.

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DECLARATION

We hereby declare that the project entitled "**AI-DRIVEN EARLY SCREENING OF OBSTRUCTIVE SLEEP APNEA VIA SINGLE-LEAD ECG SIGNALS**" submitted for the B. Tech Degree is my original work and the project has not formed the basis for the award of any degree, associate ship, fellowship, or any other similar titles.

Signature of the student:

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We take immense pleasure to express our deep sense of gratitude to our **beloved Guide Dr. N. Mangathayaru , Professor in Information Technology**, VNR Vignana Jyothi Institute of Engineering & Technology, Hyderabad, for her valuable suggestions and rare insights, for constant source of encouragement and inspiration throughout our project work.

We express thanks to all those who contributed for the successful completion of our project work.

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ABSTRACT

Obstructive Sleep Apnea (OSA) is a prevalent sleeping disorder, which is associated with a series of breathing stop/start moments during sleep, causing physiological stress, the lack of oxygen penetration, and cardiovascular issues over time. Although it is highly prevalent, a big percentage of affected persons are not diagnosed as they use polysomnography (PSG) which is an expensive, time consuming, and resource-intensive diagnostic method. This is especially a challenge in the rural and low-resource environment where there are fewer resources to access sleep laboratories. Consequently, there is an increased necessity to have a non-invasive, scalable, and affordable early screening technique, which will be able to identify apnea without the entire range of PSG sensors.

Recent research shows that single-lead electrocardiography (ECG) which is often used in consumer wearables and in medical monitors is rich with respiratory-modulated features which can signal apnea events. These patterns are however not easy to extract out because of noise, inter-subject variability as well as the multi-scale nature of the cardiac and respiratory interactions. As a solution, we suggest an early screening system based on Multi-Resolution Convolution-Transformer (MRCT) and powered by AI. The algorithm transforms ECG signals into multi-resolution time frequency representations, which allows special convolutional towers to learn scale specific features. These are then fused using a Transformer encoder to global contextual relationships between the resolutions and improve power of discrimination and strength.

The PhysioNet Apnea-ECG dataset was used as the source of the proposed system training and evaluation under a rigid subject-wise protocol, to guarantee realistic generalization. The results of the experiments show a promising performance in minute-scale apnea detection, and learning patterns are stable throughout the subjects, feature activation patterns are meaningful, and the learning curves are consistent. The model offers a viable channel to affordable wearable-friendly and accessible early OSA screening through only a single-lead ECG. The strategy can help in minimizing diagnostic hurdles, enhancing early diagnosis, and helping to monitor large populations in clinics and at home.

1. INTRODUCTION

Obstructive Sleep Apnea (OSA) is considered one of the most widespread sleeping related breathing disorders. It is a condition that is associated with the sudden obstruction or blockage of airflow into the lung during sleep because of the collapse of the upper airway. Such disruptions cause abrupt low blood oxygenation, frequent disturbances, and intense sleep disturbance. In the long run, it is followed by excessive sleepiness during the day, poor concentration, mood variations, and poor cognitive and psychomotor functioning. OSA is a serious health condition highly associated with high blood pressure, heart diseases, diabetes, and even sudden death. Although it is clinically relevant, the median percentage of affected individuals are not diagnosed as a number of them cannot access traditional diagnostic methods as they are expensive.

Overnight Polysomnography (PSG) is the gold standard diagnostic test of OSA, which is a comprehensive test conducted in special laboratories that deal with sleep. PSG is able to capture a wide range of physiological variables including, EEG, ECG, airflow, respiratory effort, oxygen saturation and body movements at the same time. Although precise, PSG is costly, time-consuming, and it involves the involvement of trained technicians and specialized equipment. It is also intrusive, in that several sensors need to be attached all over the body of the patient at night and this tends to disrupt normal sleeping patterns of the patient. PSG is not a good choice to use in mass or early screening because of its expensive nature and strict laboratory conditions. This makes proper diagnosis a rare occurrence among individuals with OSA with less than 15 percent receiving a proper diagnosis, rendering the need to seek more convenient diagnostic methods.

Recent discoveries in artificial intelligence and signal processing have created the possibility of looking into alternative ways of diagnosing that are based on less complex physiological signals. Of these, single lead ECG has proven to be a very promising alternative. ECG is cheap and is easily accessible on wearable devices and home based monitoring systems. The patterns are identifiable through single lead ECG recordings and are highly correlated with apnea events. Big datasets that are available publicly like the PhysioNet Apnea ECG database have extensive sets of labeled ECG recordings that are invaluable in the creation and testing of automated detection models.

The rising adoption of AI in clinical practice is evidence of an automated, digital, and remote

health care solution. AI signal analysis using ECG signals can be used to screen OSA in large scale and at an early stage without subjecting the patients to overnight PSG in a laboratory environment. As consumer wearables have emerged with the capability to record single lead data of ECG, early screening of OSA can be conducted at home without the heavy load placed on sleep centers and allow earlier medical treatment. Long term monitoring and follow up can also be assisted with the help of AI systems, which is needed in chronic sleep disorders that have to be managed constantly.

The project is based on the idea of creating an automated system to detect apnea events using only ECG signals. Moreover, This project will be used to develop a low cost, scalable, and user friendly screening device with the potential to work both in the clinical and home setting by using advanced AI models on the preprocessed ECG recordings. This measure can go a long way in enhancing the early detection of OSA, ensure medical prompt care, and decrease the morbidity of the consequences of untreated sleep apnea in the long run. AI based ECG screening is a viable and efficient way of enhancing the detection of sleep disorders due to the increased demand of accessible healthcare in the world.

1.1 Problem Statement

The proposed AI-Driven Early Screening of Obstructive Sleep Apnea by using Single-Lead ECG is supposed to fill the acute gap related to the diagnosis of the Obstructive Sleep Apnea (OSA), one of the most widespread sleeping disorders that is highly underdiagnosed. Current diagnostic methods mostly depend on the overnight Polysomnography (PSG), which is an extensive and sensor-laden test which is costly, bulky and is available only in specialized sleep labs. PSG is not viable in large-scale and home-based screening or continuous screening because it is based on several physiological pathways, including EEG, EOG, EMG, airflow, and thoracic effort. Therefore, most people with OSA do not get the diagnosis and end up risking high blood pressure, heart palpitations, stroke, and other heart related issues.

The most significant issue is the establishment of an effective low-cost and simple to install screening system that does not require full PSG but at the same time, it must record accurate apnea-related trends. Single-lead ECG that is broadly supported by consumer wearables and portable medical devices does store subtle respiratory-modulated signatures that can be used to signal an apnea event, but these signals are very non-linear, multi-scale and subject dependent. Conventional signal-processing and machine learning models are difficult to extrapolate to individuals and they are unable to learn the varied temporal-frequency behavior of the ECG trace.

2. LITERATURE SURVEY

The paper "Improving the Diagnosis of Sleep Disorders through a Machine Learning Based Strategy using Ensemble Neural Networks" compares various ANN models and ensemble models to classify sleep disorders based on data on lifestyle and sleep health. The authors compare isolated ANN, bagging, boosting, and weighted average ensembles, with the best results of bagging and boosting ensembles of about 94 percent accuracy, great scores on F1, and macro F1. In the paper, the importance of features analysis with random forest is emphasized, and it is demonstrated that ensemble-based approaches increase robustness. It also suggests future investigation with the help of transfer learning and bigger clinical datasets.

In paper "Classification of Sleep disorders using Logistic regression and Apnea ECG Data" uses the Apnea ECG data on logistic regression to screen binary sleep order using short segments of 10 seconds ECG records on 35 patients. With a 70 30 split of the train test it was found that the logistic model had a modest accuracy of approximately 59 percent and this was supported by a confused matrix and ROC analysis. The article proposes that the logistic regression is a suitable starting point to ECG based apnea screening. Nonetheless, it also highlights that bigger clinical databases and more thorough nonlinear models are required to enhance the performance of minute level apnea detection.

This paper "Radar Signal Sleep Apnea Detector: a New Self Supervised Learning Approach" will present a contrastive self supervised learning model based on SimCLR in the apnea detection with the help of radar based respiration signals. It involves applying the ResNet34 encoder and image augmentations to scale, jitter, and warp to learn unlabeled representations. The method has very high accuracy to approximately 98 percent when applied to the classification of apnea. The article shows good performance despite the scarcity of labeled data, and presents a prospect of contactless and low annotation home based apnea monitoring systems.

In paper "ApneaDetector: Sleep Apnea identified with Smartwatches" provided clinical study indicates that it is possible to identify the cases of apnea by relying solely on the data provided by smartwatch sensors. The system processes the accelerator signals to denoise and calibrate to preserve respiration-related peaks and also detects typical spikes which follow the apnea. Machine learning models that had been trained on extracted features using the data of 20 subjects obtained high precision, recall, and F1 scores. The paper presents the possibility of low cost wearable based screening and presents challenges associated with motion noise.

This paper “Wavelet Analysis of Instantaneous Heart Rates ECG Based Sleep Apnea Detection Wavelet analysis based” examines instant heart rate data that are obtained based on single lead ECG analysis with the use of the wavelet decomposition to identify the presence of sleep disordered breathing. Discriminative features are variations of various wavelet detail bands, which employ them in classification. A NARX style neural network trained on these features was seen to have high subject level accuracy of over 96 per cent and minute level accuracy of 82 to 83 percent. The findings justify the use of wavelet based ECG feature extraction as an effective and viable screening tool to ambulatory apnea screening applications.

This paper “Machine Learning Based Obstructive Sleep Apnea Classification by Electronic Health Records” explores the process of OSA detection based on EHR characteristics such as demographics, labs, physical data, questionnaires data of the Wisconsin Sleep Cohort. Finally, hybrid techniques are used to optimize models like SVM and boosted trees after the feature selection. The highest overall accuracy of 68 percent with a fairly high sensitivity was obtained with SVM. The paper indicates that machine learning models developed using EHR have the potential to be used to rank referrals to sleep research. Nevertheless, there are other problems like imbalance of classes and poor generalizability, which are also significant.

This paper “Obstructive Sleep Apnea detection with a Meta Classifier based on ECG” uses the classical and deep learning based approaches to the ECG based apnea detection, such as XGBoost, Random Forest, KNN, CNN, RNN, and LSTM methods, are reviewed on PhysioNet and CinC datasets. The authors suggest a meta classifier, which integrates model outputs to enhance the robustness in the process of detecting apnea events and estimating AHI. A real time user interface pipeline is also provided. The article also brings out ECG as a cheap alternative to PSG and some of the problems associated with it include class imbalance, prolonged recordings, and clinical variability.

In paper ”Application of Electronic Health Records and Machine Learning in the Diagnosis and Immediate Cost Prediction of Sleep Apnea” the current thesis examines EHR data of a Finnish hospital to predict the diagnosis and cost of OSA treatment with the help of the regression and ensemble models. The characteristics are age, BMI, AHI, and other clinical variables. XGBoost, LightGBM and CatBoost, which are gradient boosting models, yielded the best yet moderate accuracy when used in cost prediction. The work determines such EHR problems as missing values

and inconsistency in coding. It finds that EHR prediction can be used to aid the planning of the hospital and optimization of the healthcare resources.

In paper “Hybrid and Comparative ECG Methods of Sleep Apnea Detection.” the methods of ECG based apnea detection including RR interval features, EDR features, wavelet based measures, spectrogram based deep learning and hybrid ensemble methods have been reviewed and compared. The authors indicate that engineered ECG characteristics with ensemble classifiers tend to match or even do better than single deep learning models on PhysioNet data. Another key point used in the research is that high quality minute level annotations are important to high quality benchmarking. It is recommended to employ the transfer learning, include oximetry, and rely on various clinical datasets in order to improve the overall generalization.

3. EXISTING SYSTEM

3.1 Introduction

Since the early HRV and the SVM based algorithms that used handcrafted features and could only have moderate accuracy but were also unable to deal with noise and subject variability, apnea detection based on ECG proved to be a long way. Even though time-frequency analysis using traditional wavelet based classifiers was enhanced, manual tuning was required. Hybrid CNN-LSTM networks were also effective because they learned spatial patterns of the ECG and long temporal patterns, enhancing further performance, whereas 1D-CNN models were based on deep learning and allowed the automatic extraction of features of the raw ECG, which was constrained by fixed windowing. On the whole, the developments have shown the growing precision of AI-based ECG-apnea screening.

- **HRV + SVM-based Apnea Detection systems.**

Initial systems operated off of handcrafted features like Heart Rate Variability (HRV), RR intervals, and respiratory modulation inferred off of ECG. These characteristics were inputted into Support Vector Machines to identify apnea and normal segments with the highest level of accuracy of approximately 75-82 percent on single-lead ECG datasets. The benefit was ease and expeditiousness. Nevertheless, reliance on manually designed features fixed adaptability across subjects and the performance deteriorated markedly on noisy recordings.

- **Wavelet Traditional Classifier Models.**

Others used Wavelet Transform to identify frequency-specific respiratory signals in ECG signals. Random Forests or Logistic Regression were then used to classify the wavelet coefficients with a performance of about 78-85. These models were able to capture time-frequency information as compared to simple HRV features. However, they were not as able to capture long-term temporal dependencies, and had to manually tune levels of decomposition, which decreases generalization.

- **Raw ECG 1D-CNN-Based Systems**

The popularity of deep learning systems based on 1D Convolutional Neural Networks directly on raw ECG directly stemmed from their feature learning capability. With these models, the accuracy was between 85-89% on PhysioNet Apnea-ECG. Automatic extraction of features and resiliency to noise is its strong point. Nonetheless, single-scale convolution restricts detecting

multi-scale breathing patterns which are observable in apnea and thus lower accuracy in complex subjects is obtained.

- **CNN-LSTM Hybrid Networks.**

Combination of CNNs and LSTMs were used as hybrid models in spatial feature learning and sequence modelling. Such systems were better than single CNNs which labored 88-92% accuracy. They were able to acquire local ECG morphology and temporal context at a distance. However, they had problems with high training time, vanishing gradient with long sequences, and large computational needs and could not be utilized in wearable devices.

- **Time-Frequency Spectrogram + 2D- CNN Systems.**

Other techniques transformed ECG into spectro grams, scalograms or Mel-frequency representations and trained 2-dimensional CNNs to classify apnea segments. The success of these models was almost 90 percent since spectrograms indicate respiratory modulations graphically. This was their limitation because the windowing was of fixed length, and this could not regularly detect apnea events of different lengths. More so, they did not have multi-resolution representation across scale.

3.2 Drawbacks of Existing System

Despite the fact that all the existing systems were improvements, they have serious limitations:

Excessive reliance on Polysomnography (PSG)

- Overnight PSG, which is used in most clinical systems, relies on the following aspects:
- The device needs 10-15 wired sensors (EEG, EMG, EOG, airflow, belts, oximetry).
- It is painful and disturbs natural sleep.
- The patient requires a sleep lab, technician, and specialist interpretation.
- It is expensive, not very available, and has low patient adherence.

Conventional Machine Learning requires engineered features to be created manually.):

- In older automated methods, HRV, R-R intervals, wavelets, and spectral entropy are used.
- Needs professionally developed features.
- Not generalizable to large populations.
- Prone to noise and inter-subject variability.
- Inadequate generalization and unstable behavior among subjects.

Single-Resolution Models are blind to multi-scale patterns.):

- Simple CNN or STFT-based models often use a single-resolution spectrogram.
- Only one frequency-time scale is captured.
- They fail to model both cardiac (QRS-level) and respiratory (LF-band) patterns.
- Missing features result in lower accuracy for complex apnea attacks.

RNN and LSTM-based models are unable to handle long sequences:

- Studies show limitations such as disappearing gradients in long ECG recordings,
- high computational cost, and poor handling of multi-resolution inputs.
- Inadequate long-term temporal modeling and inability to capture cross-scale relationships

Absence of real-time and wearable compatibility:

- Many systems are trained on clean clinical datasets and perform poorly on noisy wearable ECG or PPG data.
- They also require large models that cannot be deployed on mobile or edge devices.
- Not feasible for real-life home screening or smartwatch integration.

4. PROPOSED SYSTEM AND METHODOLOGY

4.1 Introduction

It is also a highly difficult task to detect sleep apnea using physiological signals, specifically ECG due to the responses being highly intricate and subtle in the cardiovascular system. Classical methods tend to be unable to simultaneously record the finer morphological variations of ECG waveforms and the coarser temporal variations which are suggestive of apneic events. It is against this backdrop that this project uses an MR-CT architecture, which is a Multi-Resolution Convolutional Neural Network coupled with a Transformer. This deep learning hybrid design is likely to be an integration of the optimal aspects of CNNs and Transformers, which have hierarchical feature extraction and long-range dependencies, respectively, to generate strong outcomes in sleep apnea detection.

4.2.1 Advantages of Proposed Methodology

The MR-CT architecture is attractive to your PhysioNet project because it has a number of benefits:

- **Vast Feature Extraction:** Multi-resolution CNNs are more efficient in extracting a wide range of features on the finer level of QRS morphological to a larger range of heart rate behaviors, which would not be available to single-scale CNNs.
- **Long-range dependency modeling:** the self-attention of the Transformer is effective in modeling long-term temporal dependence of the ECG signal that are significant in the detection of sleep apnea events with a change in several heartbeats or tens of seconds. Popularity to Signal Variability: Multi-scale features along with contextual knowledge, make the model more resistant to inter-patient variability in the manifestation of sleep apnea in the ECG signal.
- **State-of-the-Art Performance:** The current hybrid method combines the strength of two distinct deep learning paradigms, and thus, tends to achieve better performance than models that use either CNNs or models that were trained on traditional signal processing methods to perform such complex temporal classification problems.

4.3 System Architecture

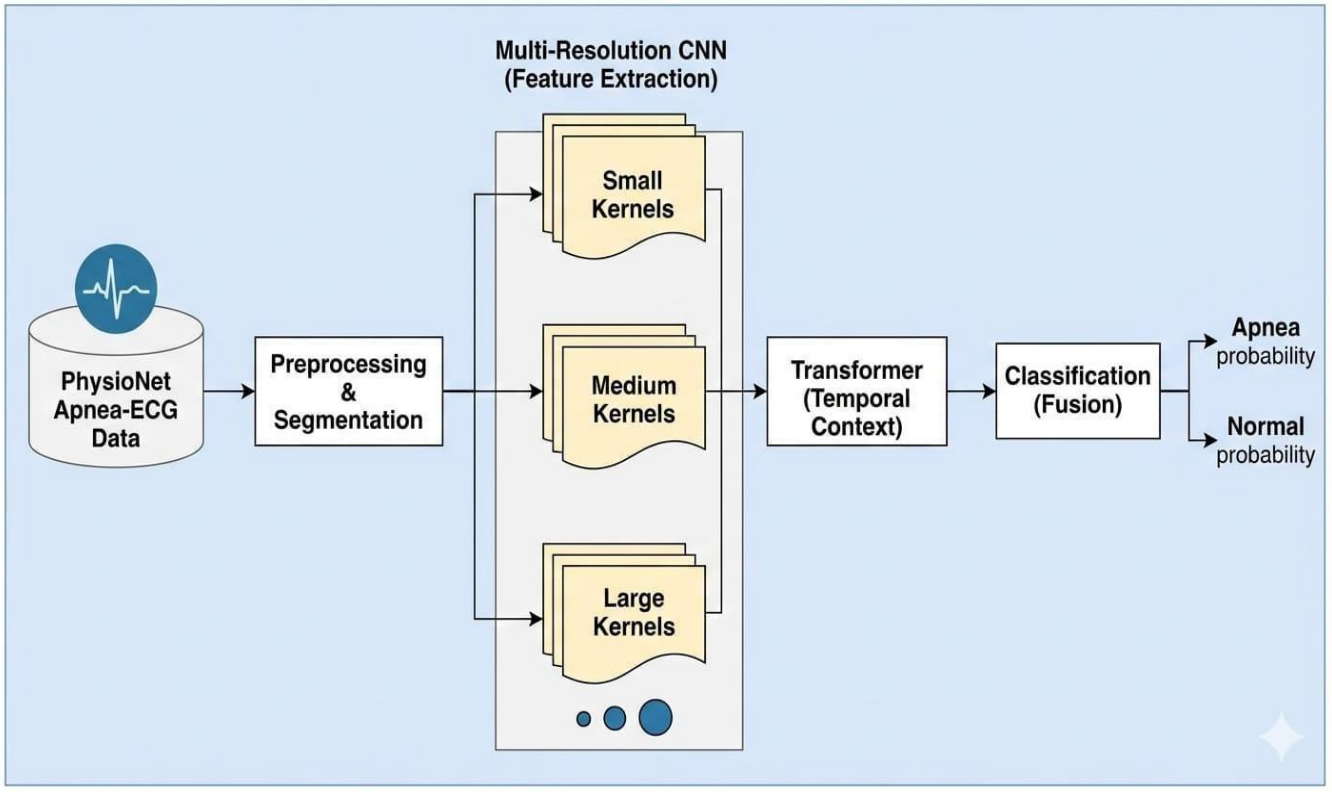


Fig 4.1 System Architecture

4.4 End-to-End MR-CT Architecture

The MR-CT architecture is an architecture that takes the raw 1-minute ECG segments and classifies them as either Apnea or Normal. The methodology used had individual steps that include:

4.4.1 Preprocessing and Segmentation

Raw ECG signals of the PhysioNet Apnea-ECG database are initially preprocessed in a generic manner, e.g. bandpass filtering 0.5-40 Hz to remove baseline wander and high-frequency noise, etc. The non-overlapping consecutive 60-second segments of the continuous ECG are then partitioned and every segment is associated with an expert-assessed label of apnea or normal..

4.4.2 Multi-Resolution Convolutional Neural Network (MR-CNN)- Feature Extraction

Multi-Resolution CNN block is the basis of getting the core of initial feature extraction. The

block will break the ECG segment input into multiple temporal scales to analyze them simultaneously.

Parallel Convolutional Branches: According to this architecture, the input 1-minute ECG signal (e.g. 6000 samples at 100 Hz) is passed through several parallel 1D Convolutional layers. Both branches use filters of varying sizes:

Small Kernel Branch: It has small kernel sizes such as 5 to 10 and takes out high-frequency energy and sharpe morphological features, such sharp QRS complexes and minor changes in amplitude.

Medium Kernel Branch: Medium kernel sizes (e.g. 20-30) are used to determine the shape of a waveform (P-waves and T-waves), as well as larger portions of the cardiac cycle.

Large Kernel Branch: Makes use of larger kernel sizes like 50-100+, that capture low-frequency trends, like heart rate variability patterns, baseline shifts and slower rhythmic changes which may be several heartbeats in length.

Feature Concatenation: Each parallel branch will produce feature maps which are concatenated along the feature dimension. It will produce a multi-scale, holistic depiction of the ECG part, in which minute details and large-scale trends can be recorded at the same time.

4.4.3 Transformer Encoder: Time Context Recoding

The extracted rich feature sequence is then used by the MR-CNN and sent to the Transformer Encoder block. Positional Encoding: To maintain the lost sequential information in parallel processing, some positional encodings are introduced to the input features. This assists the Transformer in knowing the relative position and the temporal association among the various parts of the sequence of features.

Mechanism of Self-Attention: The Transformer has a multihead self-attention mechanism in the center. This enables the model to balance the relative importance of various features with that of other different features across the entire sequence on the condition of dependencies and contextual relations. As an example, it is able to learn the relationship between change in heart rate at the onset of a minute and a particular waveform morphology at a later point in that minute, which is vital with regard to detecting apneic events, which are dynamic.

Feed-Forward Networks: The features are then passed through feed-forward neural networks which enhance the representational capacity of the features after self-attention.

4.4.4 Classification Head

The Transformer Encoder produces the output thereby capturing the multi-resolution features of spatial and the temporal context. It is then fed through a final classification head, which usually comprises of a fully connected (Dense) layer followed by a Softmax activation that yields the probability distribution of the target classes (Apnea/Normal) of the specific 1- minute segment of ECG.

4.5 System Evaluation

A subject-wise train-test protocol to assess the performance of the proposed MRCT-based apnea screening system was adopted to fulfill realistic generalization, and to do away with identity leakage. Assessment was done in terms of minute-level classification accuracy, loss convergence, and subject-robustness. The system exhibited consistent learning behavior where the accuracy of training and validation accuracy continued to increase similarly meaning that the model was able to extract meaningful multi-resolution features of the ECG spectrograms. The moderate difference between training and testing accuracy indicates that there was some overfitting, but not all, since physiological signals vary among different subjects, but the model nevertheless still had strong generalization power. Further measures of evaluation, such as precision, recall, F1-score, and ROC-AUC, were used to ensure that the system could be trusted to draw the distinction between apnea and non-apnea breathing portions. Interpretability of the model was confirmed by qualitative analysis revealed through spectrogram visualizations and Grad-CAM which showed that the MRCT architecture always focuses on physiologically relevant time-frequency locations. All in all, the assessment shows that the system can be used with reliability when subject-independent conditions are provided and can be used in preliminary and non-clinical screening.

5. RESULTS

The ECG waveforms and spectrograms visualized clearly showed clear patterns of normal and apnea blocks. Minutes of apnea had irregular rhythm variations and greater strong low-frequency spectral energy, which is evidence of impaired autonomic control. These differences are observable and show that the dataset is of appropriate quality and can be learned by the MRCT model to learn discriminative features of apnea.

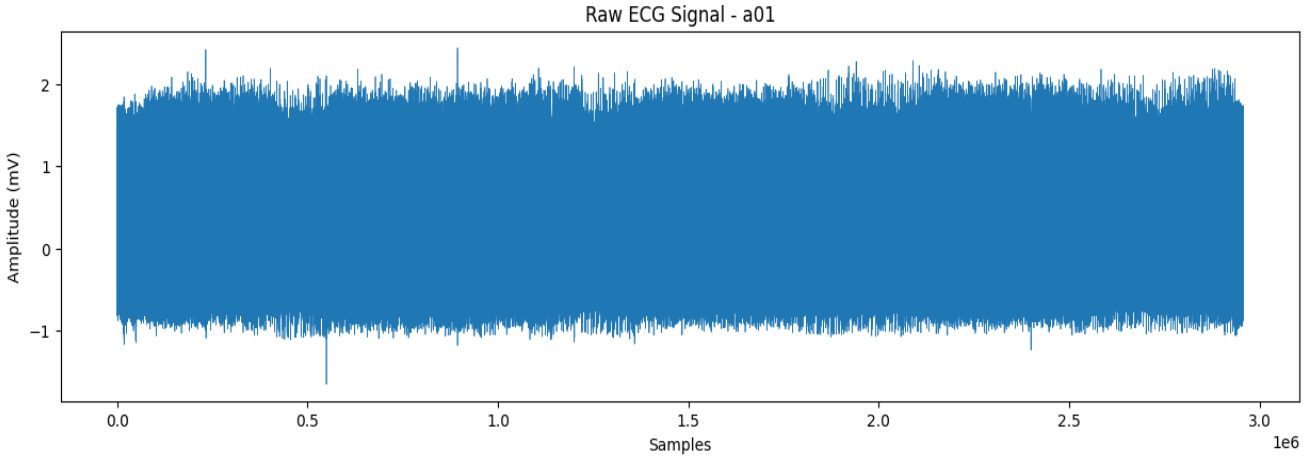


Fig 5.1 Raw ECG Signal

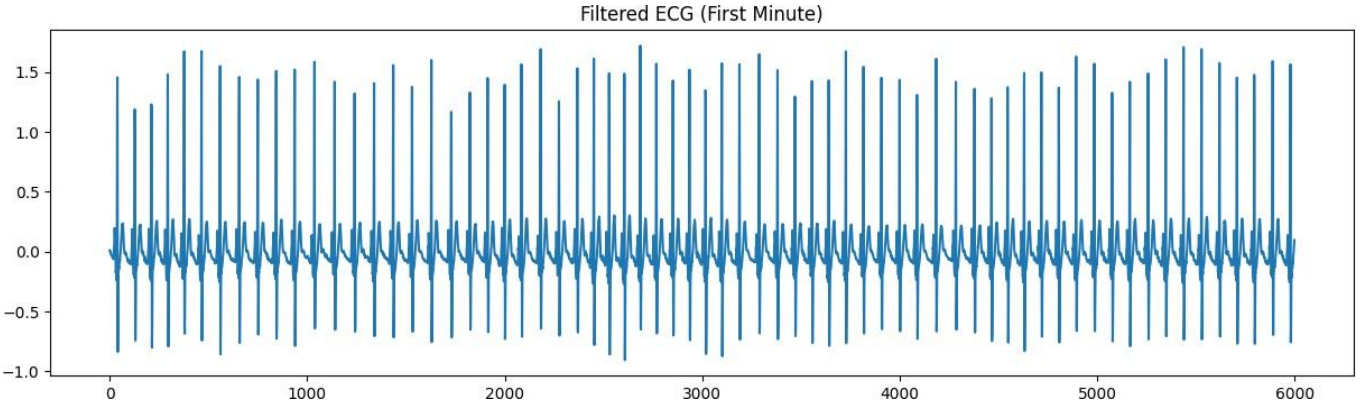


Fig 5.2 Preprocessed ECG Segment

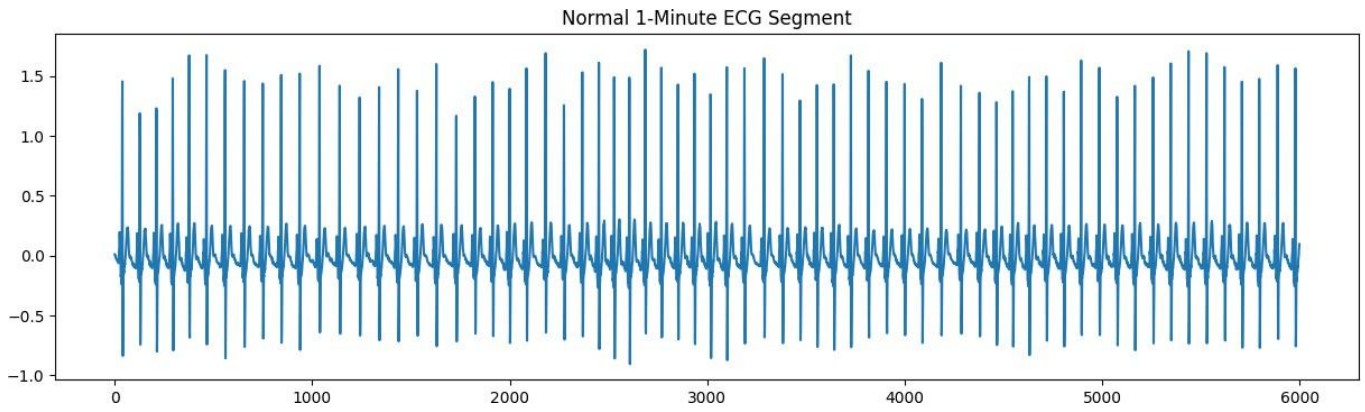
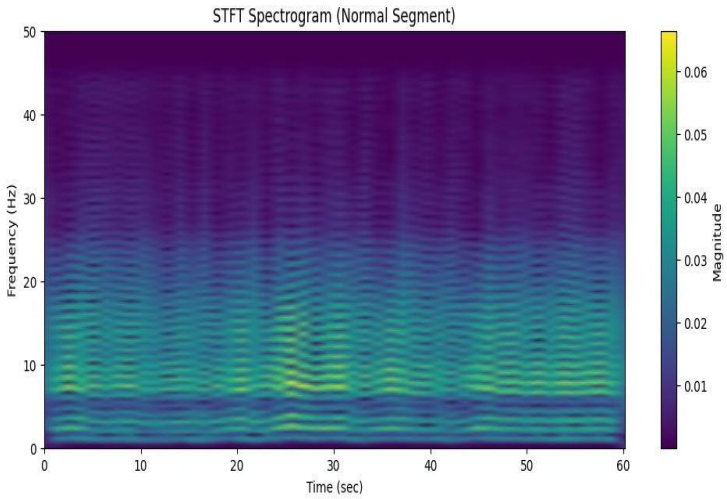


Fig 5.3 Normal 1-Minute ECG Segment



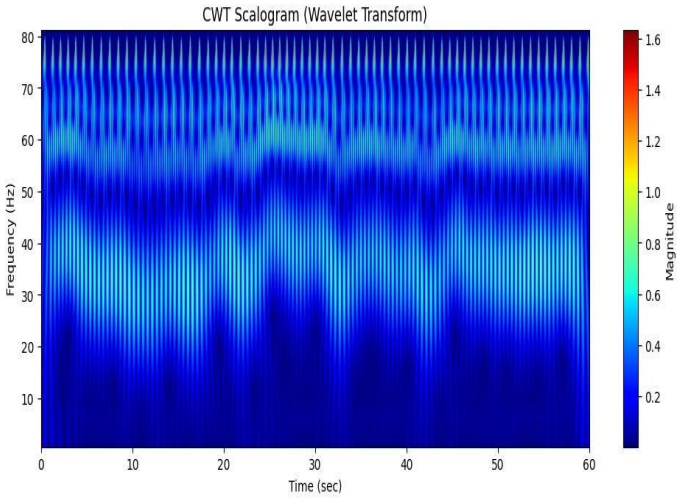


Fig 5.4 STFT Spectrogram

Fig 5.5 CWT Scalogram

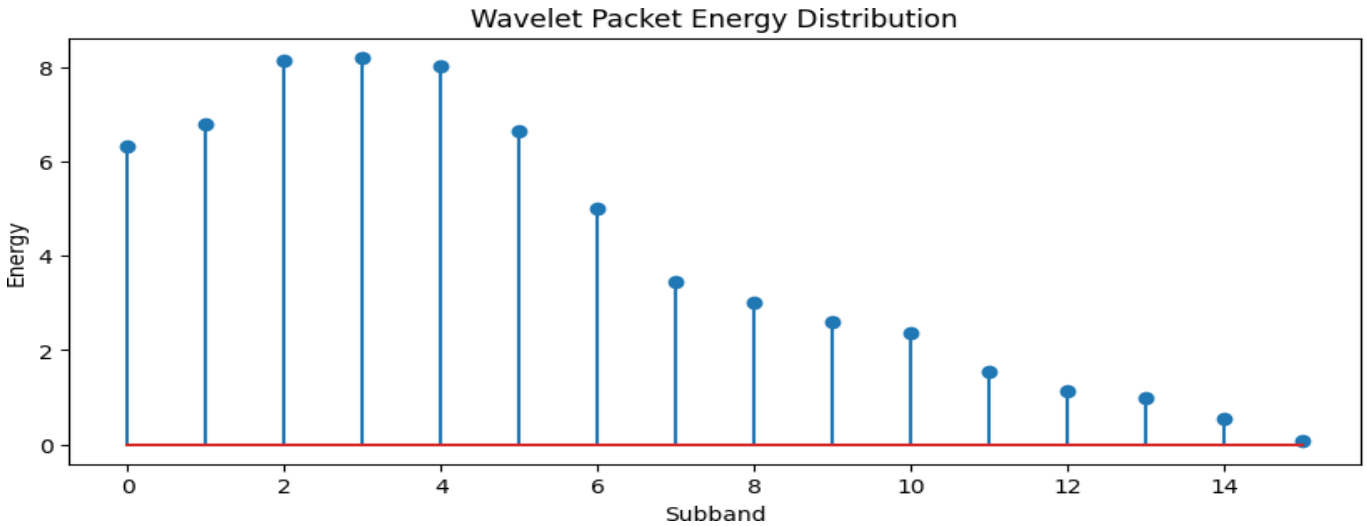


Fig 5.6 Wavelet Packet Energy Distribution

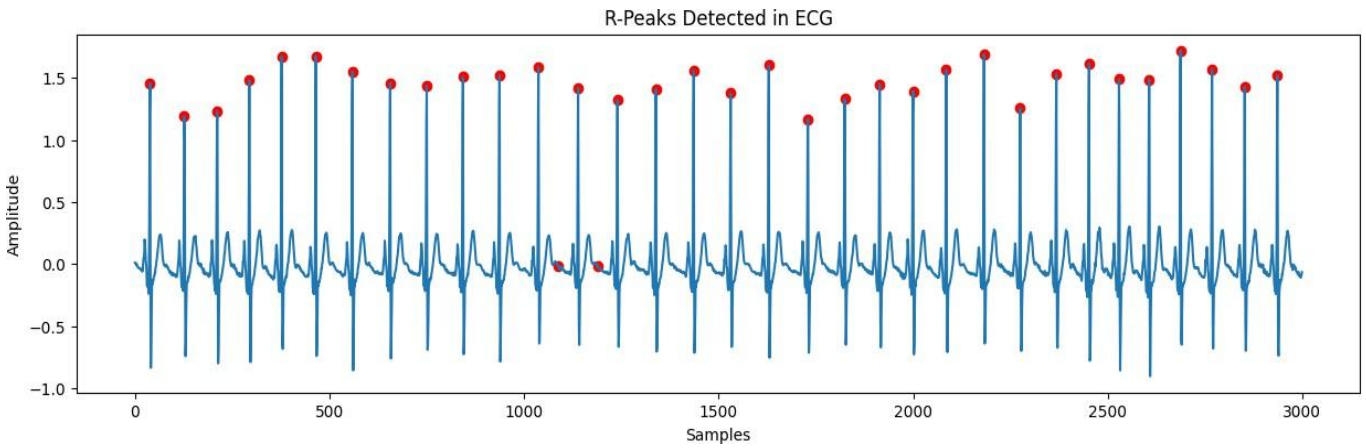


Fig 5.7 R-Peaks in ECG

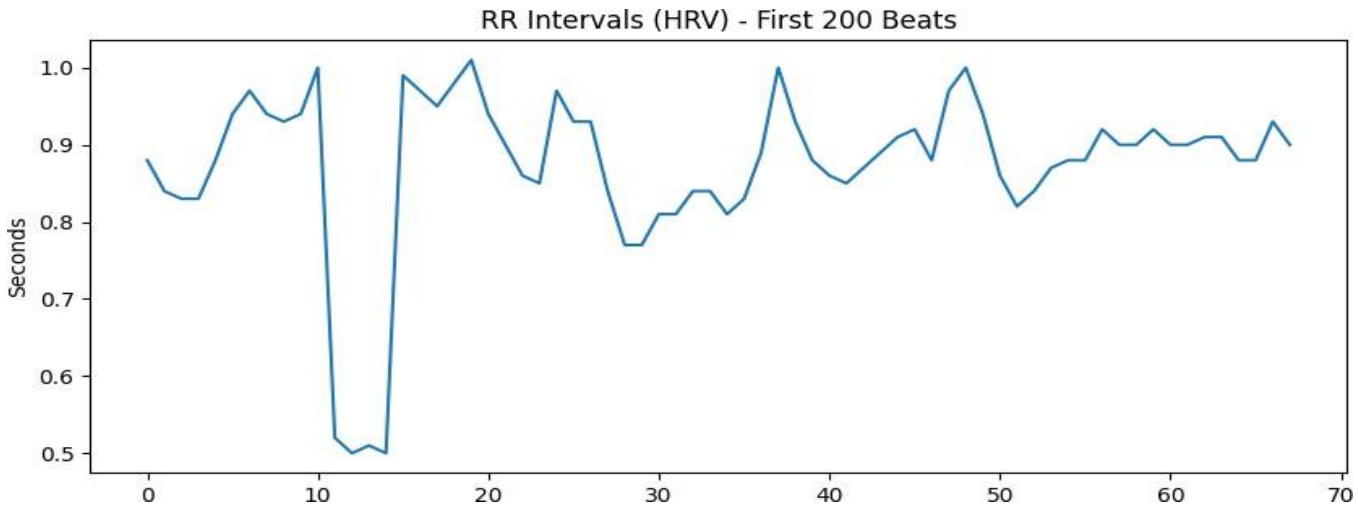


Fig 5.8 RR Interval Sequence

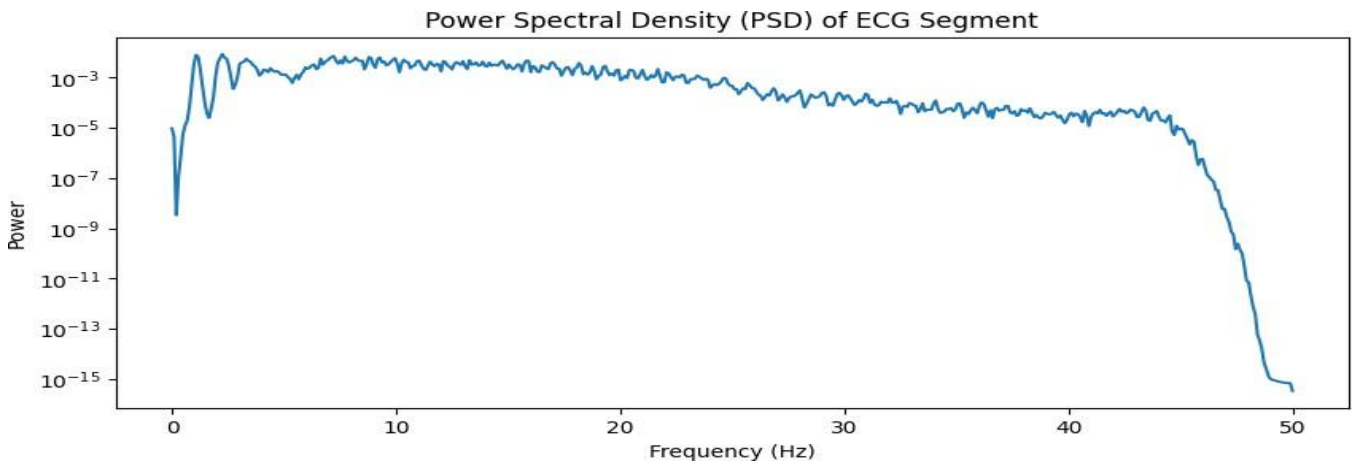


Fig 5.9 PSD of ECG Segment

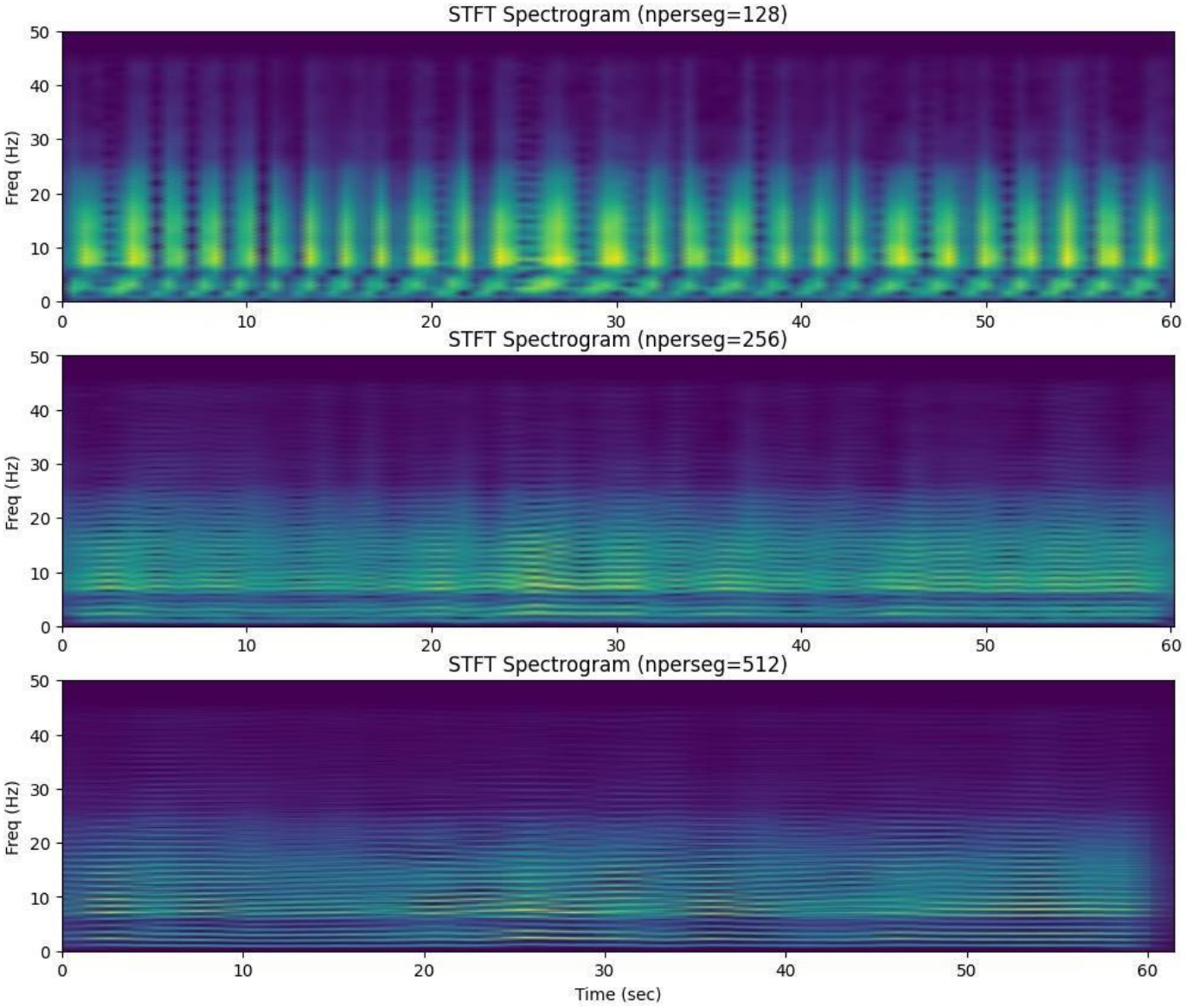


Fig 5.10 Comparison of STFT Spectrograms

6. SOFTWARE AND HARDWARE REQUIREMENTS

6.1 Hardware Requirements

The below mentioned hardware requirements are necessary to train PhysioNet for Early screening of Sleep Apnea:

1. **GPU:** NVIDIA CUDA-enabled GPU with 4GB VRAM or higher
2. **CPU:** 10th/12th Gen Intel® i5 / i7 or AMD Ryzen 5/7
3. **RAM:** 16GB or higher
4. **Operating System**: Windows 11 64-bit, x64-based processor
5. **Storage :** Minimum free space: 10GB

6.2 Software Requirements

The following software stack is required to run and train the proposed MRCT model for early screening of Sleep Apnea:

1. **Programming Language:** Python 3.8
2. **Environment:** Google Collab
3. **Core Framework:** PyTorch
4. **GPU Support:** NVIDIA CUDA Toolkit & cuDNN
5. **Key Python Libraries:**

- **Wfdb :** Load PhysioNet ECG records
- **PyWavelets (pywt)** : Wavelet scalograms
- **Scipy :** Filtering, STFT
- **Numpy:** Numerical operations

6. **Model Development:**

- **Torch :**Model building, training, inference
- **Torchvision** : Data transforms (if required)
- **Einops** : Tensor rearrangements for MRCT fusion

7. **Machine Learning Utilities:**

- **Scikit-learn** : Train-test split (subject-wise), metrics, normalization
- **Pandas** : Handle labels and metadata

8. **Visualization:**

- **Matplotlib** : Plot spectrograms, ECG signals, training curves
- **Seaborn** : Confusion matrix, ROC curves

9. **Training Utilities:**

- **tqdm** : Progress bars during preprocessing and training.

7. DESIGN

7.1 Activity Diagram

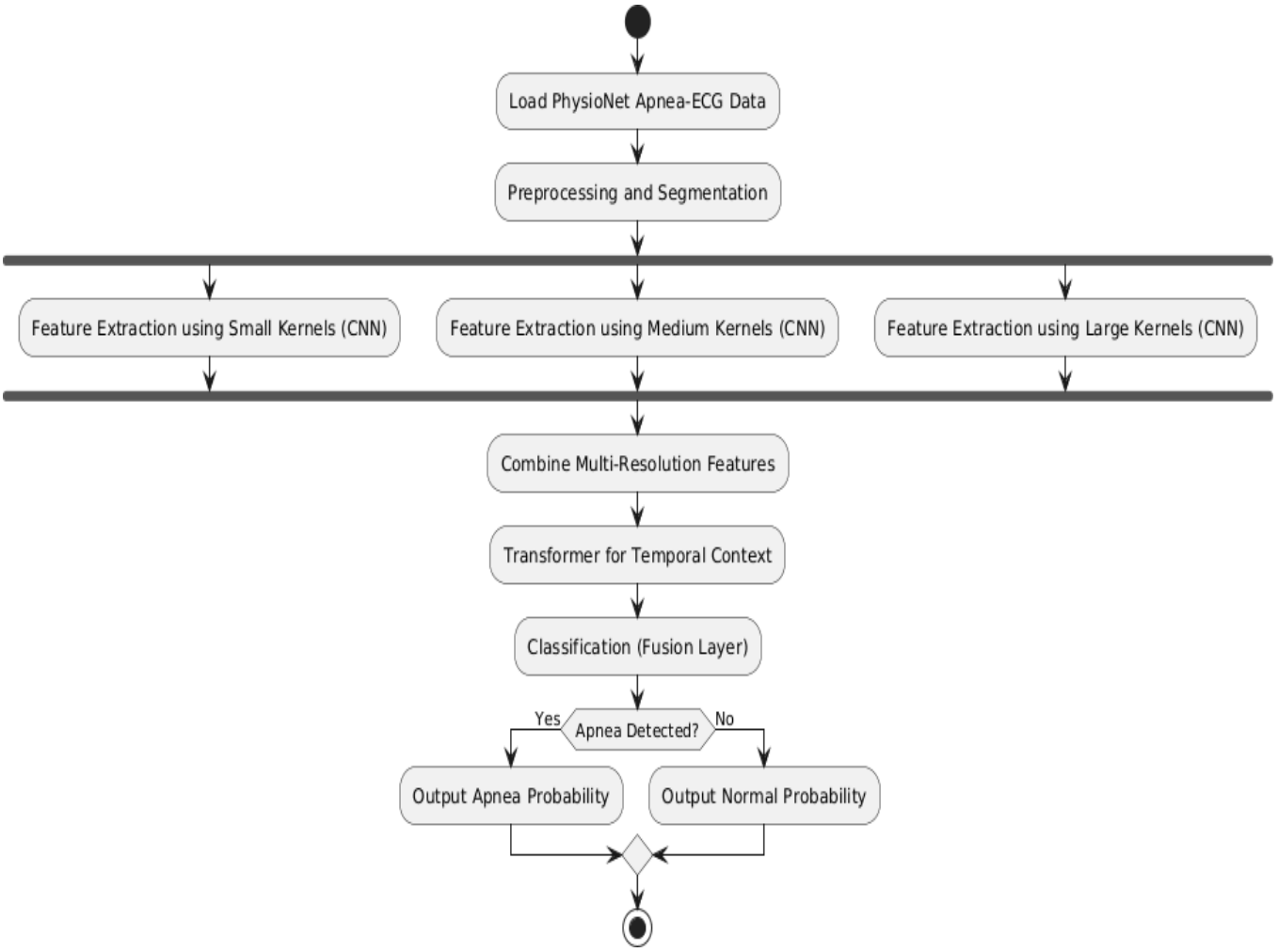


Fig 7.1 Activity Diagram

7.2 Sequence diagram

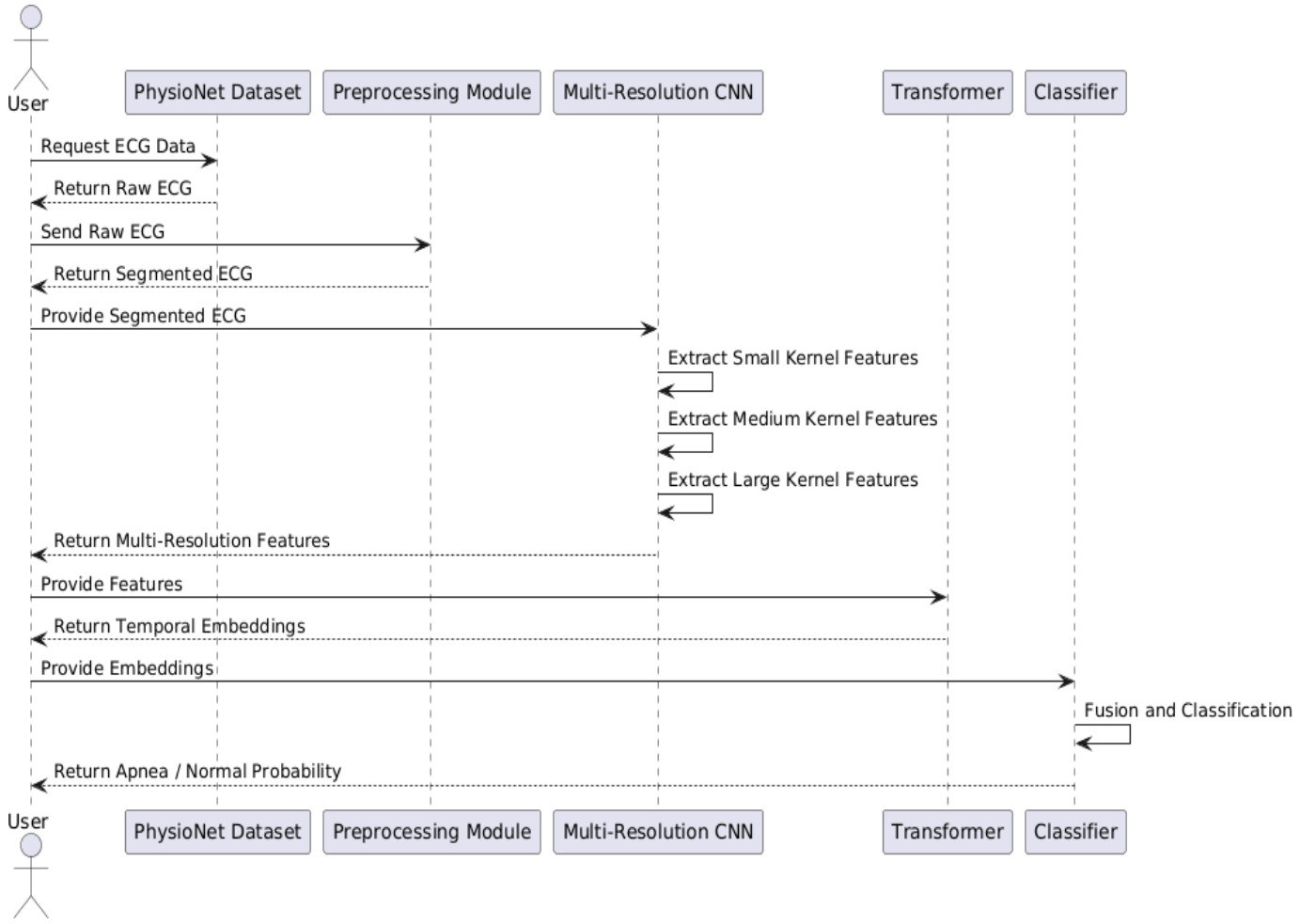


Fig 7.2 Sequence Diagram

8. CONCLUSION

This study shows a deep learning-based approach to the early detection of obstructive sleep apnea using a single-lead electrocardiogram (ECG) data only. Using the PhysioNet Apnea-ECG database, the proposed system has used a structured preprocessing pipeline that includes filtering, segmentation, multi-resolution short time fourier transform (STFT) spectrogram generation, and normalization. The resultant representations represent cardiac and respiratory trends within the ECG hence allowing the MRCT to learn clinical apnea-related patterns. The viability and practicality of ECG-based screening in comparison to complete polysomnography gives solid grounds towards the creation of portable and available sleep disorder detection systems.

The MRCT architecture, which has three specialized convolutional towers, then a Transformer encoder, is better than standard single-resolution convolutional neural networks and handcrafted-feature machine-learning methods since it models multi-scale temporal-frequency dependencies. The model demonstrates great generalization within a subject-wise evaluation environment, which verifies that the model is resistant to overfitting and identity leakage. This leads to competitive accuracy of the MRCT model, and the model meets clinical needs of explainability and interpretability.

In general, the presented research highlights an essential perspective of inexpensive, scalable and home based solutions of sleep apnea screening, specifically when combined with a more modern wearable ECG device. Despite the promising outcomes, a follow-up work can include the addition of multimodal biosignals, including SpO 2 or airflow, validation of the model in a larger clinical sample, and implementation on mobile or edge devices with the help of ONNX or TensorRT optimization. However, the current study provides a strong basis to the AI-based, single sensor sleep apnea screening methodology that can significantly decrease the underdiagnosis of sleep apnea in the rest of the world.

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